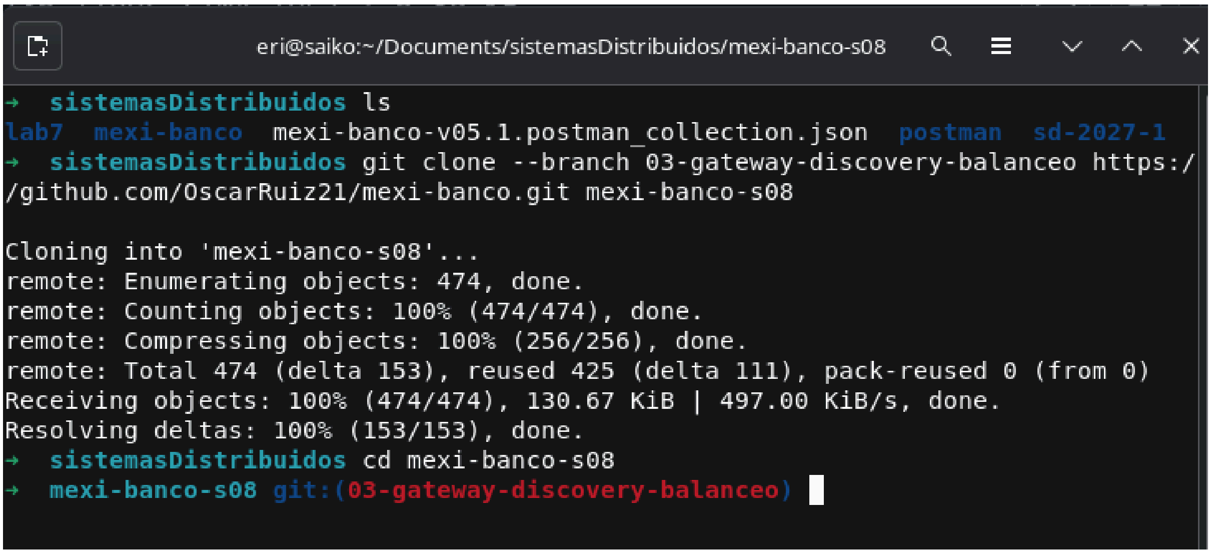
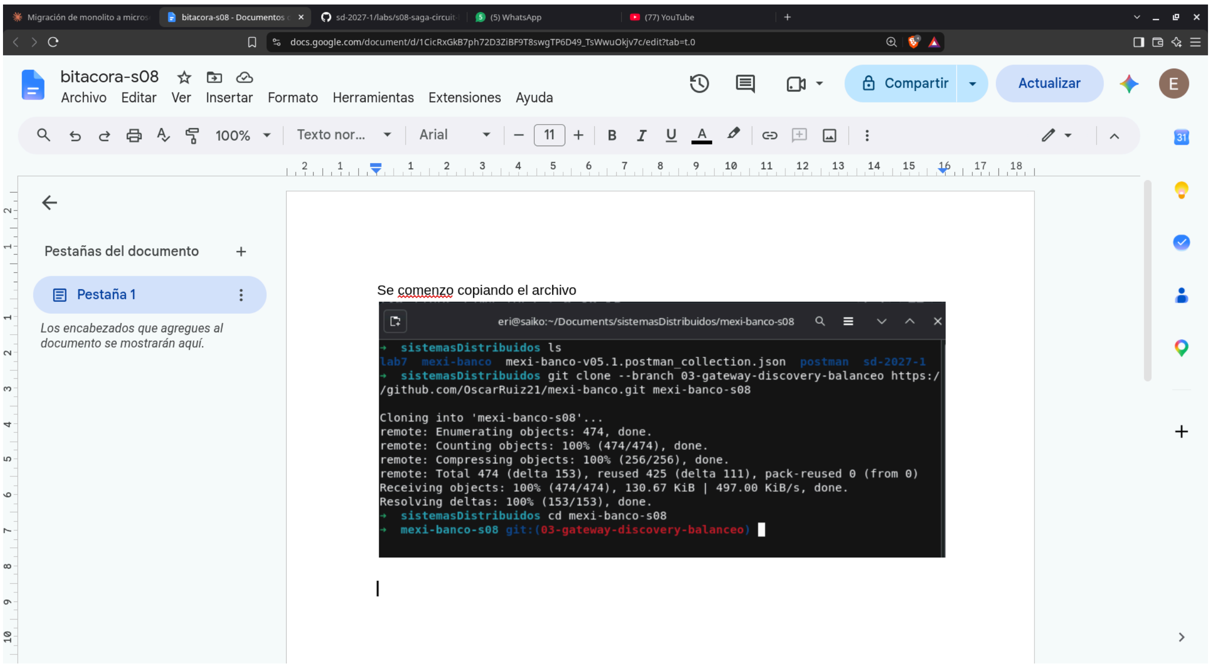
Erick Nava Santiago
Sistemas Distribuidos - Lab 08

Se comenzó copiando el archivo


eri@saiko:~/Documents/sistemasDistribuidos/mexi-banco-s08

```
➜  sistemasDistribuidos ls
lab7  mexi-banco  mexi-banco-v05.1.postman_collection.json  postman  sd-2027-1
➜  sistemasDistribuidos git clone --branch 03-gateway-discovery-balanceo https:/
/github.com/OscarRuiz21/mexi-banco.git mexi-banco-s08

Cloning into 'mexi-banco-s08'...
remote: Enumerating objects: 474, done.
remote: Counting objects: 100% (474/474), done.
remote: Compressing objects: 100% (256/256), done.
remote: Total 474 (delta 153), reused 425 (delta 111), pack-reused 0 (from 0)
Receiving objects: 100% (474/474), 130.67 KiB | 497.00 KiB/s, done.
Resolving deltas: 100% (153/153), done.
➜  sistemasDistribuidos cd mexi-banco-s08
➜  mexi-banco-s08 git:(03-gateway-discovery-balanceo) 
```

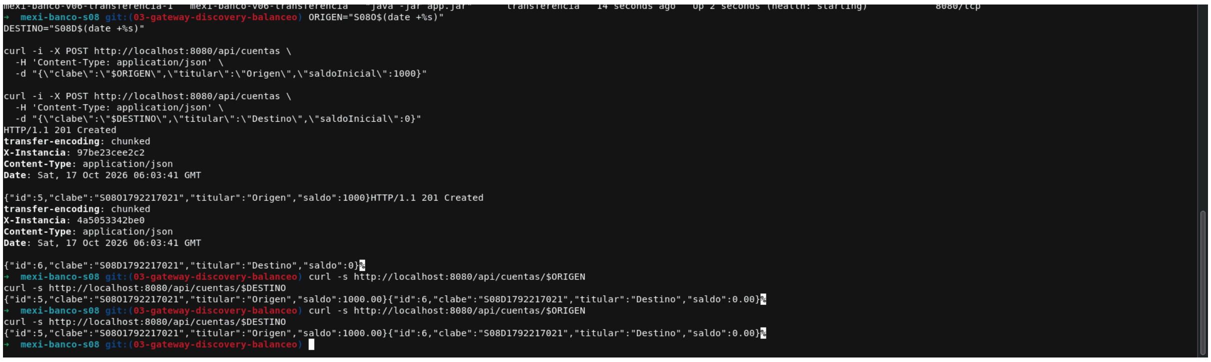
Se visualizo


Migración de monolito a micros
bitacora-s08 - Documentos
sd-2027-1/labs/s08-saga-circuit
(5) WhatsApp
(77) YouTube
docs.google.com/document/d/1CicRxGkB7ph72D3ZiBF9T8swgTP6D49_TsWwuOkjv7c/edit?tab=t.0
bitacora-s08
Archivo
Editar
Ver
Insertar
Formato
Herramientas
Extensiones
Ayuda
Compartir
Actualizar
100%
Texto nor...
Arial
11
Pestañas del documento
Pestaña 1
Los encabezados que agregues al documento se mostrarán aquí.

Se comenzo copiando el archivo

```
➜  sistemasDistribuidos ls
lab7  mexi-banco  mexi-banco-v05.1.postman_collection.json  postman  sd-2027-1
➜  sistemasDistribuidos git clone --branch 03-gateway-discovery-balanceo https:/
/github.com/OscarRuiz21/mexi-banco.git mexi-banco-s08

Cloning into 'mexi-banco-s08'...
remote: Enumerating objects: 474, done.
remote: Counting objects: 100% (474/474), done.
remote: Compressing objects: 100% (256/256), done.
remote: Total 474 (delta 153), reused 425 (delta 111), pack-reused 0 (from 0)
Receiving objects: 100% (474/474), 130.67 KiB | 497.00 KiB/s, done.
Resolving deltas: 100% (153/153), done.
➜  sistemasDistribuidos cd mexi-banco-s08
➜  mexi-banco-s08 git:(03-gateway-discovery-balanceo) 
```

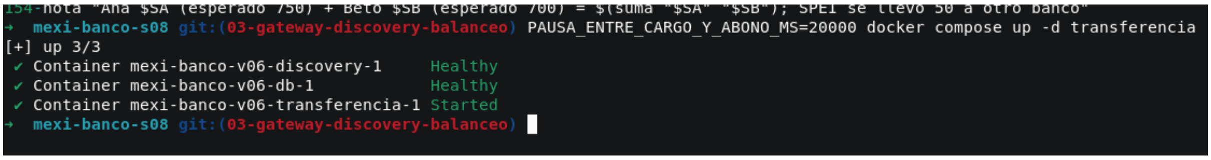
1. Se crearon los servicios



```
mexi-banco-v06-transferencia-1   mexi-banco-v06-transferencia   "java -jar app.jar"   transferencia   14 seconds ago   Up 2 seconds (health: starting)   8080/tcp
➜  mexi-banco-s08 git:(03-gateway-discovery-balanceo) ORIGEN="S08O$(date +%s)"
DESTINO="S08D$(date +%s)"

curl -i -X POST http://localhost:8080/api/cuentas \
  -H 'Content-Type: application/json' \
  -d "{\"clabe\":\"$ORIGEN\",\"titular\":\"Origen\",\"saldoInicial\":1000}"

curl -i -X POST http://localhost:8080/api/cuentas \
  -H 'Content-Type: application/json' \
  -d "{\"clabe\":\"$DESTINO\",\"titular\":\"Destino\",\"saldoInicial\":0}"
HTTP/1.1 201 Created
transfer-encoding: chunked
X-Instancia: 97be23cee2c2
Content-Type: application/json
Date: Sat, 17 Oct 2026 06:03:41 GMT

{"id":5,"clabe":"S08O1792217021","titular":"Origen","saldo":1000}HTTP/1.1 201 Created
transfer-encoding: chunked
X-Instancia: 4a5053342be0
Content-Type: application/json
Date: Sat, 17 Oct 2026 06:03:41 GMT

{"id":6,"clabe":"S08D1792217021","titular":"Destino","saldo":0}%
➜  mexi-banco-s08 git:(03-gateway-discovery-balanceo) curl -s http://localhost:8080/api/cuentas/$ORIGEN
curl -s http://localhost:8080/api/cuentas/$DESTINO
{"id":5,"clabe":"S08O1792217021","titular":"Origen","saldo":1000.00}{"id":6,"clabe":"S08D1792217021","titular":"Destino","saldo":0.00}%
➜  mexi-banco-s08 git:(03-gateway-discovery-balanceo) curl -s http://localhost:8080/api/cuentas/$ORIGEN
curl -s http://localhost:8080/api/cuentas/$DESTINO
{"id":5,"clabe":"S08O1792217021","titular":"Origen","saldo":1000.00}{"id":6,"clabe":"S08D1792217021","titular":"Destino","saldo":0.00}%
➜  mexi-banco-s08 git:(03-gateway-discovery-balanceo) 
```

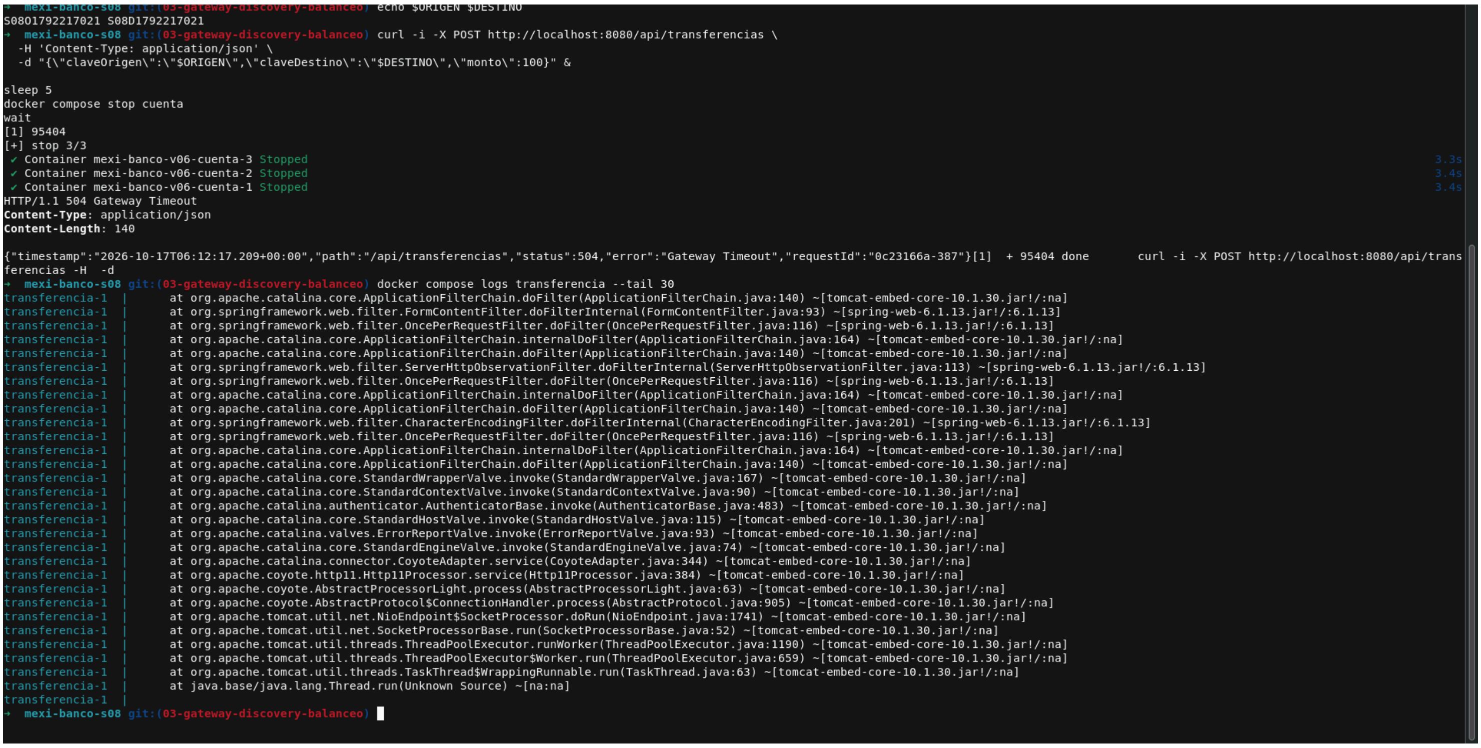
Erick Nava Santiago
Sistemas Distribuidos - Lab 08

2. Se levantó transferencia con una pausa



```
154-nota "Ana $SA (esperado 750) + Beto $SB (esperado 700) = $(suma "$SA" "$SB"); SPEI se llevo 50 a otro banco"
→  mexi-banco-s08 git:(03-gateway-discovery-balanceo) PAUSA_ENTRE_CARGO_Y_ABONO_MS=20000 docker compose up -d transferencia
[+] up 3/3
 ✔ Container mexi-banco-v06-discovery-1     Healthy
 ✔ Container mexi-banco-v06-db-1            Healthy
 ✔ Container mexi-banco-v06-transferencia-1 Started
→  mexi-banco-s08 git:(03-gateway-discovery-balanceo) █
```

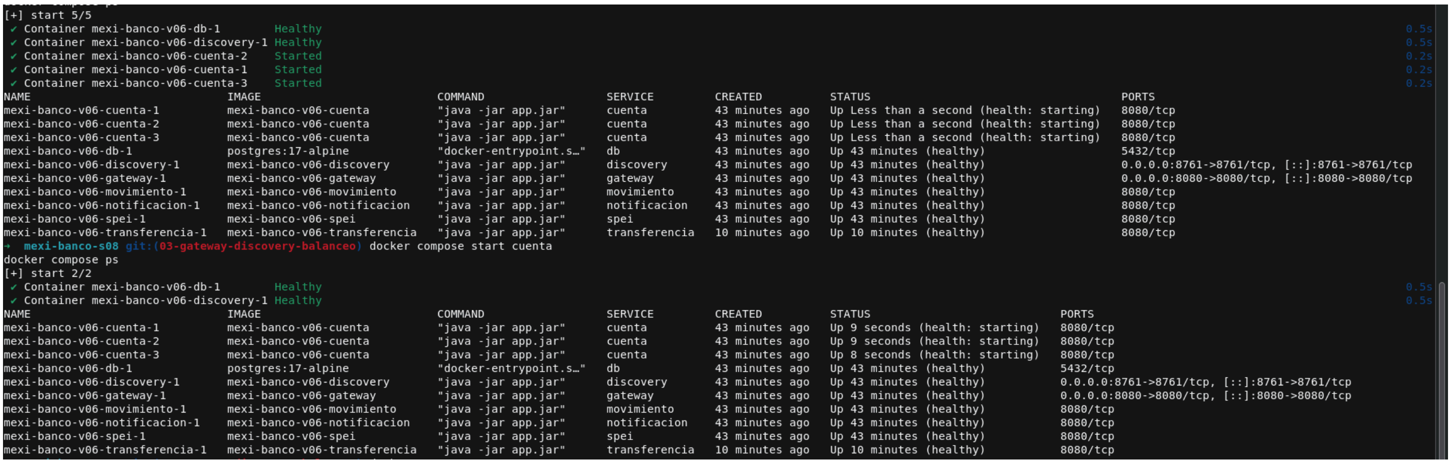
3. Luego se lanzo una transferencia y se apagó cuenta. Se recibe un error



```
→  mexi-banco-s08 git:(03-gateway-discovery-balanceo) echo $ORIGEN $DESTINO
S08O1792217021 S08D1792217021
→  mexi-banco-s08 git:(03-gateway-discovery-balanceo) curl -i -X POST http://localhost:8080/api/transferencias \
  -H 'Content-Type: application/json' \
  -d "{\"claveOrigen\":\"$ORIGEN\",\"claveDestino\":\"$DESTINO\",\"monto\":100}" &

sleep 5
docker compose stop cuenta
wait
[1] 95404
[+] stop 3/3
 ✔ Container mexi-banco-v06-cuenta-3 Stopped                                                                                                                                   3.3s
 ✔ Container mexi-banco-v06-cuenta-2 Stopped                                                                                                                                   3.4s
 ✔ Container mexi-banco-v06-cuenta-1 Stopped                                                                                                                                   3.4s
HTTP/1.1 504 Gateway Timeout
Content-Type: application/json
Content-Length: 140

{"timestamp":"2026-10-17T06:12:17.209+00:00","path":"/api/transferencias","status":504,"error":"Gateway Timeout","requestId":"0c23166a-387"}[1]  + 95404 done       curl -i -X POST http://localhost:8080/api/trans
ferencias -H  -d
→  mexi-banco-s08 git:(03-gateway-discovery-balanceo) docker compose logs transferencia --tail 30
transferencia-1  |      at org.apache.catalina.core.ApplicationFilterChain.doFilter(ApplicationFilterChain.java:140) ~[tomcat-embed-core-10.1.30.jar!/:na]
transferencia-1  |      at org.springframework.web.filter.FormContentFilter.doFilterInternal(FormContentFilter.java:93) ~[spring-web-6.1.13.jar!/:6.1.13]
transferencia-1  |      at org.springframework.web.filter.OncePerRequestFilter.doFilter(OncePerRequestFilter.java:116) ~[spring-web-6.1.13.jar!/:6.1.13]
transferencia-1  |      at org.apache.catalina.core.ApplicationFilterChain.internalDoFilter(ApplicationFilterChain.java:164) ~[tomcat-embed-core-10.1.30.jar!/:na]
transferencia-1  |      at org.apache.catalina.core.ApplicationFilterChain.doFilter(ApplicationFilterChain.java:140) ~[tomcat-embed-core-10.1.30.jar!/:na]
transferencia-1  |      at org.springframework.web.filter.ServerHttpObservationFilter.doFilterInternal(ServerHttpObservationFilter.java:113) ~[spring-web-6.1.13.jar!/:6.1.13]
transferencia-1  |      at org.springframework.web.filter.OncePerRequestFilter.doFilter(OncePerRequestFilter.java:116) ~[spring-web-6.1.13.jar!/:6.1.13]
transferencia-1  |      at org.apache.catalina.core.ApplicationFilterChain.internalDoFilter(ApplicationFilterChain.java:164) ~[tomcat-embed-core-10.1.30.jar!/:na]
transferencia-1  |      at org.apache.catalina.core.ApplicationFilterChain.doFilter(ApplicationFilterChain.java:140) ~[tomcat-embed-core-10.1.30.jar!/:na]
transferencia-1  |      at org.springframework.web.filter.CharacterEncodingFilter.doFilterInternal(CharacterEncodingFilter.java:201) ~[spring-web-6.1.13.jar!/:6.1.13]
transferencia-1  |      at org.springframework.web.filter.OncePerRequestFilter.doFilter(OncePerRequestFilter.java:116) ~[spring-web-6.1.13.jar!/:6.1.13]
transferencia-1  |      at org.apache.catalina.core.ApplicationFilterChain.internalDoFilter(ApplicationFilterChain.java:164) ~[tomcat-embed-core-10.1.30.jar!/:na]
transferencia-1  |      at org.apache.catalina.core.ApplicationFilterChain.doFilter(ApplicationFilterChain.java:140) ~[tomcat-embed-core-10.1.30.jar!/:na]
transferencia-1  |      at org.apache.catalina.core.StandardWrapperValve.invoke(StandardWrapperValve.java:167) ~[tomcat-embed-core-10.1.30.jar!/:na]
transferencia-1  |      at org.apache.catalina.core.StandardContextValve.invoke(StandardContextValve.java:90) ~[tomcat-embed-core-10.1.30.jar!/:na]
transferencia-1  |      at org.apache.catalina.authenticator.AuthenticatorBase.invoke(AuthenticatorBase.java:483) ~[tomcat-embed-core-10.1.30.jar!/:na]
transferencia-1  |      at org.apache.catalina.core.StandardHostValve.invoke(StandardHostValve.java:115) ~[tomcat-embed-core-10.1.30.jar!/:na]
transferencia-1  |      at org.apache.catalina.valves.ErrorReportValve.invoke(ErrorReportValve.java:93) ~[tomcat-embed-core-10.1.30.jar!/:na]
transferencia-1  |      at org.apache.catalina.core.StandardEngineValve.invoke(StandardEngineValve.java:74) ~[tomcat-embed-core-10.1.30.jar!/:na]
transferencia-1  |      at org.apache.catalina.connector.CoyoteAdapter.service(CoyoteAdapter.java:344) ~[tomcat-embed-core-10.1.30.jar!/:na]
transferencia-1  |      at org.apache.coyote.http11.Http11Processor.service(Http11Processor.java:384) ~[tomcat-embed-core-10.1.30.jar!/:na]
transferencia-1  |      at org.apache.coyote.AbstractProcessorLight.process(AbstractProcessorLight.java:63) ~[tomcat-embed-core-10.1.30.jar!/:na]
transferencia-1  |      at org.apache.coyote.AbstractProtocol$ConnectionHandler.process(AbstractProtocol.java:905) ~[tomcat-embed-core-10.1.30.jar!/:na]
transferencia-1  |      at org.apache.tomcat.util.net.NioEndpoint$SocketProcessor.doRun(NioEndpoint.java:1741) ~[tomcat-embed-core-10.1.30.jar!/:na]
transferencia-1  |      at org.apache.tomcat.util.net.SocketProcessorBase.run(SocketProcessorBase.java:52) ~[tomcat-embed-core-10.1.30.jar!/:na]
transferencia-1  |      at org.apache.tomcat.util.threads.ThreadPoolExecutor.runWorker(ThreadPoolExecutor.java:1190) ~[tomcat-embed-core-10.1.30.jar!/:na]
transferencia-1  |      at org.apache.tomcat.util.threads.ThreadPoolExecutor$Worker.run(ThreadPoolExecutor.java:659) ~[tomcat-embed-core-10.1.30.jar!/:na]
transferencia-1  |      at org.apache.tomcat.util.threads.TaskThread$WrappingRunnable.run(TaskThread.java:63) ~[tomcat-embed-core-10.1.30.jar!/:na]
transferencia-1  |      at java.base/java.lang.Thread.run(Unknown Source) ~[na:na]
transferencia-1  |
→  mexi-banco-s08 git:(03-gateway-discovery-balanceo) █
```

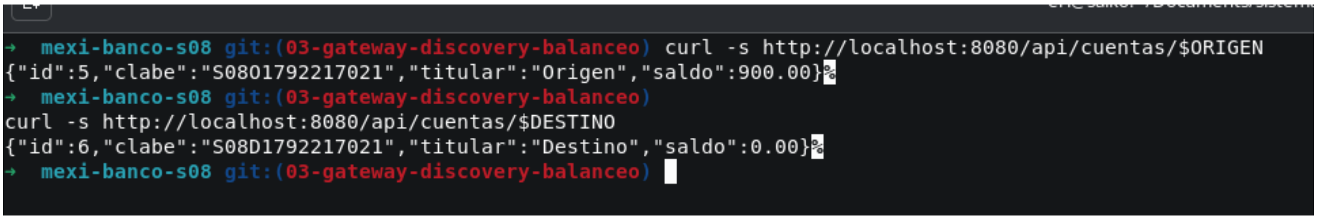
Se obtuvó un error 504 por tardar en encontrar la puerta de salida

4.



```
docker compose ps
[+] start 5/5
 ✔ Container mexi-banco-v06-db-1        Healthy                                                                                                                               0.5s
 ✔ Container mexi-banco-v06-discovery-1 Healthy                                                                                                                               0.5s
 ✔ Container mexi-banco-v06-cuenta-2    Started                                                                                                                               0.2s
 ✔ Container mexi-banco-v06-cuenta-1    Started                                                                                                                               0.2s
 ✔ Container mexi-banco-v06-cuenta-3    Started                                                                                                                               0.2s
NAME                             IMAGE                              COMMAND                    SERVICE          CREATED          STATUS                                         PORTS
mexi-banco-v06-cuenta-1          mexi-banco-v06-cuenta              "java -jar app.jar"        cuenta           43 minutes ago   Up Less than a second (health: starting)   8080/tcp
mexi-banco-v06-cuenta-2          mexi-banco-v06-cuenta              "java -jar app.jar"        cuenta           43 minutes ago   Up Less than a second (health: starting)   8080/tcp
mexi-banco-v06-cuenta-3          mexi-banco-v06-cuenta              "java -jar app.jar"        cuenta           43 minutes ago   Up Less than a second (health: starting)   8080/tcp
mexi-banco-v06-db-1              postgres:17-alpine                 "docker-entrypoint.s…"   db               43 minutes ago   Up 43 minutes (healthy)                    5432/tcp
mexi-banco-v06-discovery-1       mexi-banco-v06-discovery           "java -jar app.jar"        discovery        43 minutes ago   Up 43 minutes (healthy)                    0.0.0.0:8761->8761/tcp, [::]:8761->8761/tcp
mexi-banco-v06-gateway-1         mexi-banco-v06-gateway             "java -jar app.jar"        gateway          43 minutes ago   Up 43 minutes (healthy)                    0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp
mexi-banco-v06-movimiento-1      mexi-banco-v06-movimiento          "java -jar app.jar"        movimiento       43 minutes ago   Up 43 minutes (healthy)                    8080/tcp
mexi-banco-v06-notificacion-1    mexi-banco-v06-notificacion        "java -jar app.jar"        notificacion     43 minutes ago   Up 43 minutes (healthy)                    8080/tcp
mexi-banco-v06-spei-1            mexi-banco-v06-spei                "java -jar app.jar"        spei             43 minutes ago   Up 43 minutes (healthy)                    8080/tcp
mexi-banco-v06-transferencia-1   mexi-banco-v06-transferencia       "java -jar app.jar"        transferencia    10 minutes ago   Up 10 minutes (healthy)                    8080/tcp
→  mexi-banco-s08 git:(03-gateway-discovery-balanceo) docker compose start cuenta
docker compose ps
[+] start 2/2
 ✔ Container mexi-banco-v06-db-1        Healthy                                                                                                                               0.5s
 ✔ Container mexi-banco-v06-discovery-1 Healthy                                                                                                                               0.5s
NAME                             IMAGE                              COMMAND                    SERVICE          CREATED          STATUS                               PORTS
mexi-banco-v06-cuenta-1          mexi-banco-v06-cuenta              "java -jar app.jar"        cuenta           43 minutes ago   Up 9 seconds (health: starting)   8080/tcp
mexi-banco-v06-cuenta-2          mexi-banco-v06-cuenta              "java -jar app.jar"        cuenta           43 minutes ago   Up 9 seconds (health: starting)   8080/tcp
mexi-banco-v06-cuenta-3          mexi-banco-v06-cuenta              "java -jar app.jar"        cuenta           43 minutes ago   Up 8 seconds (health: starting)   8080/tcp
mexi-banco-v06-db-1              postgres:17-alpine                 "docker-entrypoint.s…"   db               43 minutes ago   Up 43 minutes (healthy)           5432/tcp
mexi-banco-v06-discovery-1       mexi-banco-v06-discovery           "java -jar app.jar"        discovery        43 minutes ago   Up 43 minutes (healthy)           0.0.0.0:8761->8761/tcp, [::]:8761->8761/tcp
mexi-banco-v06-gateway-1         mexi-banco-v06-gateway             "java -jar app.jar"        gateway          43 minutes ago   Up 43 minutes (healthy)           0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp
mexi-banco-v06-movimiento-1      mexi-banco-v06-movimiento          "java -jar app.jar"        movimiento       43 minutes ago   Up 43 minutes (healthy)           8080/tcp
mexi-banco-v06-notificacion-1    mexi-banco-v06-notificacion        "java -jar app.jar"        notificacion     43 minutes ago   Up 43 minutes (healthy)           8080/tcp
mexi-banco-v06-spei-1            mexi-banco-v06-spei                "java -jar app.jar"        spei             43 minutes ago   Up 43 minutes (healthy)           8080/tcp
mexi-banco-v06-transferencia-1   mexi-banco-v06-transferencia       "java -jar app.jar"        transferencia    10 minutes ago   Up 10 minutes (healthy)           8080/tcp
```

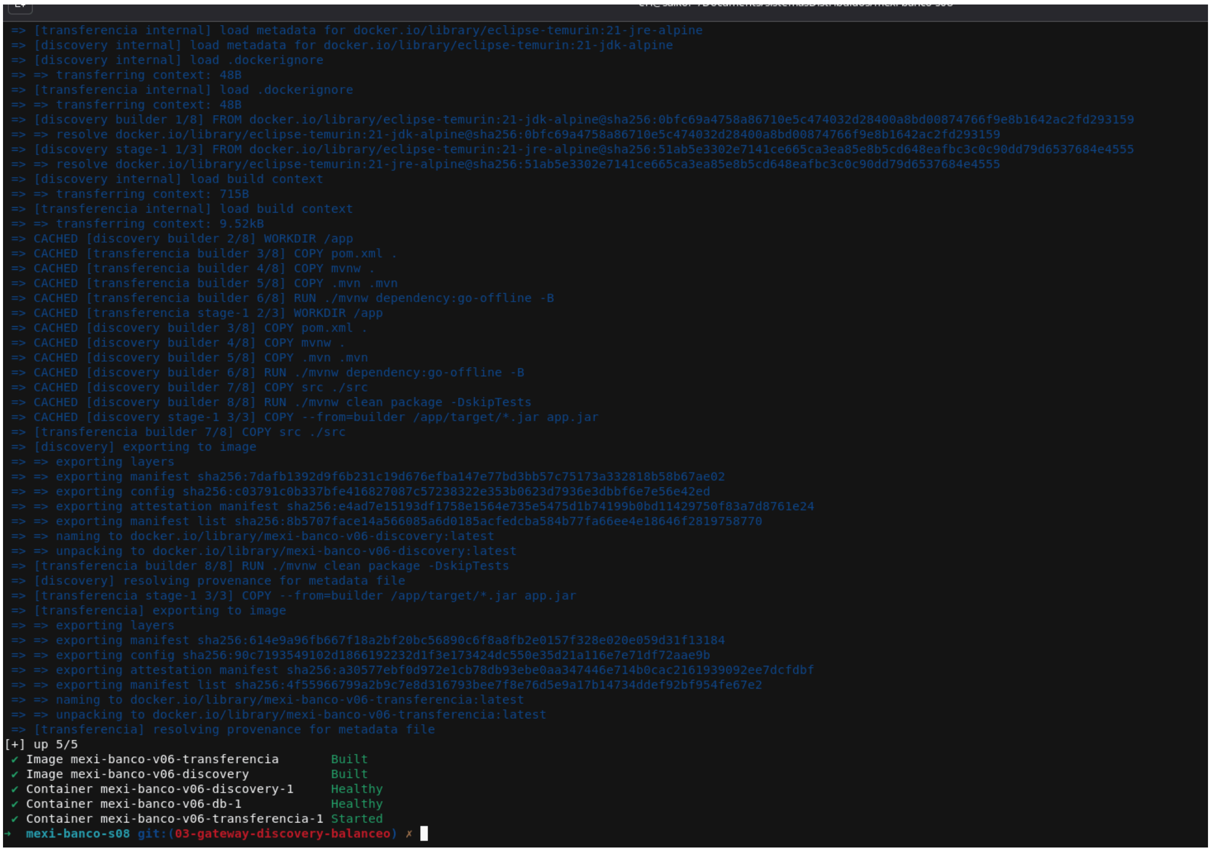
Erick Nava Santiago
Sistemas Distribuidos - Lab 08



```
→  mexi-banco-s08 git:(03-gateway-discovery-balanceo) curl -s http://localhost:8080/api/cuentas/$ORIGEN
{"id":5,"clabe":"S08O1792217021","titular":"Origen","saldo":900.00}%
→  mexi-banco-s08 git:(03-gateway-discovery-balanceo)
curl -s http://localhost:8080/api/cuentas/$DESTINO
{"id":6,"clabe":"S08D1792217021","titular":"Destino","saldo":0.00}%
→  mexi-banco-s08 git:(03-gateway-discovery-balanceo)
```

Se puede ver que no llegó al destino, siendo el saldo de 0.00.

En cuanto al código se modifica. Se modificó dentro de la carpeta correspondiente a las transferencias:



```
=> [transferencia internal] load metadata for docker.io/library/eclipse-temurin:21-jre-alpine
=> [discovery internal] load metadata for docker.io/library/eclipse-temurin:21-jdk-alpine
=> [discovery internal] load .dockerignore
=> => transferring context: 48B
=> [transferencia internal] load .dockerignore
=> => transferring context: 48B
=> [discovery builder 1/8] FROM docker.io/library/eclipse-temurin:21-jdk-alpine@sha256:0bfc69a4758a86710e5c474032d28400a8bd00874766f9e8b1642ac2fd293159
=> => resolve docker.io/library/eclipse-temurin:21-jdk-alpine@sha256:0bfc69a4758a86710e5c474032d28400a8bd00874766f9e8b1642ac2fd293159
=> [discovery stage-1 1/3] FROM docker.io/library/eclipse-temurin:21-jre-alpine@sha256:51ab5e3302e7141ce665ca3ea85e8b5cd648eafbc3c0c90dd79d6537684e4555
=> => resolve docker.io/library/eclipse-temurin:21-jre-alpine@sha256:51ab5e3302e7141ce665ca3ea85e8b5cd648eafbc3c0c90dd79d6537684e4555
=> [discovery internal] load build context
=> => transferring context: 715B
=> [transferencia internal] load build context
=> => transferring context: 9.52kB
=> CACHED [discovery builder 2/8] WORKDIR /app
=> CACHED [transferencia builder 3/8] COPY pom.xml .
=> CACHED [transferencia builder 4/8] COPY mvnw .
=> CACHED [transferencia builder 5/8] COPY .mvn .mvn
=> CACHED [transferencia builder 6/8] RUN ./mvnw dependency:go-offline -B
=> CACHED [transferencia stage-1 2/3] WORKDIR /app
=> CACHED [discovery builder 3/8] COPY pom.xml .
=> CACHED [discovery builder 4/8] COPY mvnw .
=> CACHED [discovery builder 5/8] COPY .mvn .mvn
=> CACHED [discovery builder 6/8] RUN ./mvnw dependency:go-offline -B
=> CACHED [discovery builder 7/8] COPY src ./src
=> CACHED [discovery builder 8/8] RUN ./mvnw clean package -DskipTests
=> CACHED [discovery stage-1 3/3] COPY --from=builder /app/target/*.jar app.jar
=> [transferencia builder 7/8] COPY src ./src
=> [discovery] exporting to image
=> => exporting layers
=> => exporting manifest sha256:7dafb1392d9f6b231c19d676efba147e77bd3bb57c75173a332818b58b67ae02
=> => exporting config sha256:c03791c0b337bfe416827087c57238322e353b0623d7936e3dbbf6e7e56e42ed
=> => exporting attestation manifest sha256:e4ad7e15193df1758e1564e735e5475d1b74199b0bd11429750f83a7d8761e24
=> => exporting manifest list sha256:8b5707face14a566085a6d0185acfedcba584b77fa66ee4e18646f2819758770
=> => naming to docker.io/library/mexi-banco-v06-discovery:latest
=> => unpacking to docker.io/library/mexi-banco-v06-discovery:latest
=> [transferencia builder 8/8] RUN ./mvnw clean package -DskipTests
=> [discovery] resolving provenance for metadata file
=> [transferencia stage-1 3/3] COPY --from=builder /app/target/*.jar app.jar
=> [transferencia] exporting to image
=> => exporting layers
=> => exporting manifest sha256:614e9a96fb667f18a2bf20bc56890c6f8a8fb2e0157f328e020e059d31f13184
=> => exporting config sha256:90c7193549102d1866192232d1f3e173424dc550e35d21a116e7e71df72aae9b
=> => exporting attestation manifest sha256:a30577ebf0d972e1cb78db93ebe0aa347446e714b0cac2161939092ee7dcfdbf
=> => exporting manifest list sha256:4f55966799a2b9c7e8d316793bee7f8e76d5e9a17b14734ddef92bf954fe67e2
=> => naming to docker.io/library/mexi-banco-v06-transferencia:latest
=> => unpacking to docker.io/library/mexi-banco-v06-transferencia:latest
=> [transferencia] resolving provenance for metadata file
[+] up 5/5
 ✔ Image mexi-banco-v06-transferencia       Built
 ✔ Image mexi-banco-v06-discovery           Built
 ✔ Container mexi-banco-v06-discovery-1     Healthy
 ✔ Container mexi-banco-v06-db-1            Healthy
 ✔ Container mexi-banco-v06-transferencia-1 Started
→  mexi-banco-s08 git:(03-gateway-discovery-balanceo) ✗
```

- Se agregó un programa encargado de tener el catálogo de los estados posibles.

```
package mx.mexibanco.transferencia;

public enum EstadoTransferencia{
    INICIADA,
    CARGADA,
    ABONADA,
    COMPLETADA,
    COMPENSADA,
    COMPENSACION_PENDIENTE,
    FALLIDA
}
```

Erick Nava Santiago
Sistemas Distribuidos - Lab 08

- En el archivo Transferencia.java se agregaron los metodos para el retorno de parametros de la clase

```
   public Transferencia(String claveOrigen, String claveDestino,
BigDecimal monto) {
       this.claveOrigen = claveOrigen;
       this.claveDestino = claveDestino;
       this.monto = monto;
       this.fecha = Instant.now();
       this.estado = EstadoTransferencia.INICIADA;
   }
     . . .
   public EstadoTransferencia getEstado(){
       return estado;
   }

   public String getDetalle(){
       return detalle;
   }
}
```

    - En Transferencia controller:

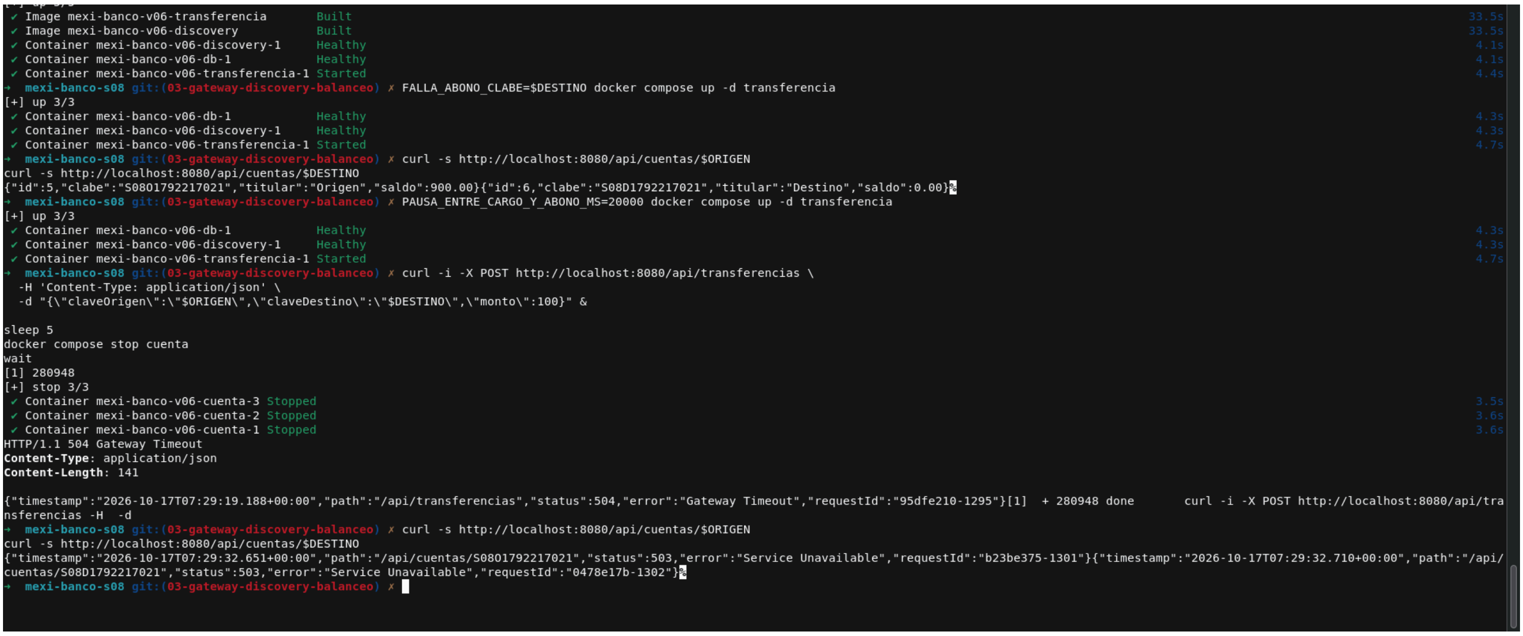
```
   /**
    * COMPLETADA --> 201
    * COMPENSADA -> 409
    * PENDIENTE -> 502
    *
    * Se agrego un remapro de los casos mostradospara el cliente:
    */

   @PostMapping
   public ResponseEntity<Transferencia> transferir(@Valid @RequestBody
SolicitudTransferencia solicitud) {
       Transferencia transferencia =
transferencias.transferir(solicitud.claveOrigen(),
solicitud.claveDestino(), solicitud.monto());
       HttpStatus estado = switch (transferencia.getEstado()) {
           case COMPLETADA -> HttpStatus.CREATED;
           case COMPENSADA -> HttpStatus.CONFLICT;
           default -> HttpStatus.BAD_GATEWAY;
       };
       return ResponseEntity.status(estado).body(transferencia);
    }
    @GetMapping("/{id}")
    public Transferencia consultar(@PathVariable Long id) {
        return transferencias.consultar(id);
    }
}
```

- En TransferenciaService se agegaron el manejo de los casos para poder gestionar la comulacion del monto y cuando hay que devolverlo. Dentro del codigo se encuentran los comentarios generales:

```
        }
        //compruba existencia
        cuentas.consultar(claveOrigen);
        cuentas.consultar(claveDestino);
        //estado inciial
        Transferencia transferencia = new Transferencia(claveOrigen,
claveDestino, monto);
        transferencias.save(transferencia);
        //cargo al origen
        CuentaRemota origen;
        try {
            origen = cuentas.cargar(claveOrigen, monto);
        } catch (RuntimeException falla) {
            //402,422
            guardar(transferencia, EstadoTransferencia.FALLIDA, "No se
pudo cargar el origen: " + falla.getMessage());
            throw falla;
        }
        guardar(transferencia, EstadoTransferencia.CARGADA, null);
        pausaDeDemo();
        //abono a destino
        CuentaRemota destino;
        try {
            fallaProvocadaDelAbono(claveDestino);
            destino = cuentas.abonar(claveDestino, monto);
        } catch (RuntimeException falla) {
            return compensar(transferencia, origen, falla);
        }
        guardar(transferencia, EstadoTransferencia.ABONADA, null);

        //asegurando la llegada del dinero
```

Erick Nava Santiago
Sistemas Distribuidos - Lab 08

```
       MovimientoRemoto cargo = movimientos.registrar(origen.clabe(),
TipoMovimiento.TRANSFERENCIA_ENVIADA,
           monto.negate(), origen.saldo(), "a " + destino.clabe());
       MovimientoRemoto abono = movimientos.registrar(destino.clabe(),
TipoMovimiento.TRANSFERENCIA_RECIBIDA,
           monto, destino.saldo(), "de " + origen.clabe());
       guardar(transferencia, EstadoTransferencia.COMPLETADA, null);
       notificaciones.notificar(cargo);
       notificaciones.notificar(abono);
       return transferencia;
       }
```

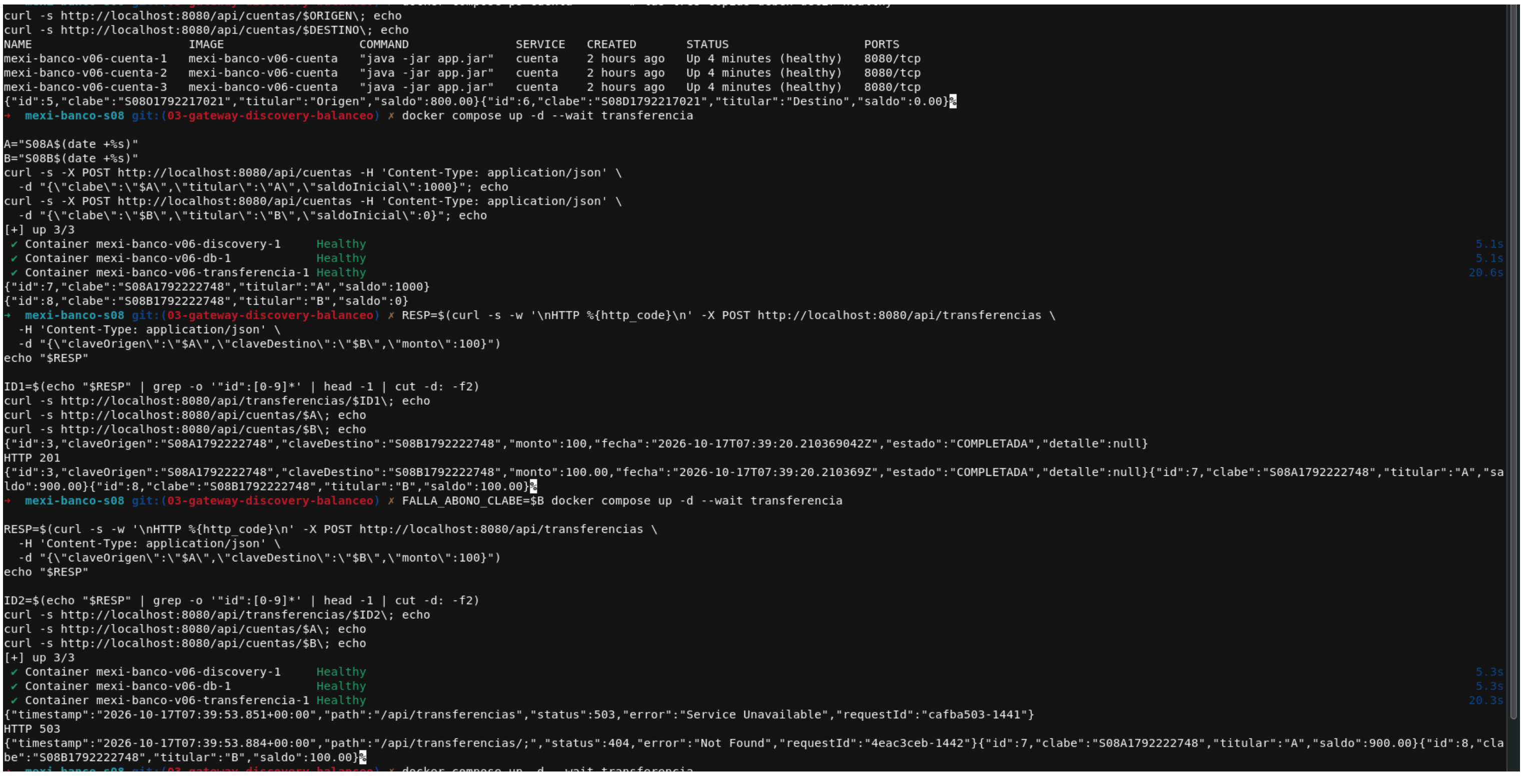


```
 ✔ Image mexi-banco-v06-transferencia       Built          33.5s
 ✔ Image mexi-banco-v06-discovery           Built          33.5s
 ✔ Container mexi-banco-v06-discovery-1     Healthy         4.1s
 ✔ Container mexi-banco-v06-db-1            Healthy         4.1s
 ✔ Container mexi-banco-v06-transferencia-1 Started         4.4s
→  mexi-banco-s08 git:(03-gateway-discovery-balanceo) ✗ FALLA_ABONO_CLABE=$DESTINO docker compose up -d transferencia
[+] up 3/3
 ✔ Container mexi-banco-v06-db-1            Healthy         4.3s
 ✔ Container mexi-banco-v06-discovery-1     Healthy         4.3s
 ✔ Container mexi-banco-v06-transferencia-1 Started         4.7s
→  mexi-banco-s08 git:(03-gateway-discovery-balanceo) ✗ curl -s http://localhost:8080/api/cuentas/$ORIGEN
curl -s http://localhost:8080/api/cuentas/$DESTINO
{"id":5,"clabe":"S08O1792217021","titular":"Origen","saldo":900.00}{"id":6,"clabe":"S08D1792217021","titular":"Destino","saldo":0.00}%
→  mexi-banco-s08 git:(03-gateway-discovery-balanceo) ✗ PAUSA_ENTRE_CARGO_Y_ABONO_MS=20000 docker compose up -d transferencia
[+] up 3/3
 ✔ Container mexi-banco-v06-db-1            Healthy         4.3s
 ✔ Container mexi-banco-v06-discovery-1     Healthy         4.3s
 ✔ Container mexi-banco-v06-transferencia-1 Started         4.7s
→  mexi-banco-s08 git:(03-gateway-discovery-balanceo) ✗ curl -i -X POST http://localhost:8080/api/transferencias \
  -H 'Content-Type: application/json' \
  -d "{\"claveOrigen\":\"$ORIGEN\",\"claveDestino\":\"$DESTINO\",\"monto\":100}" &

sleep 5
docker compose stop cuenta
wait
[1] 280948
[+] stop 3/3
 ✔ Container mexi-banco-v06-cuenta-3 Stopped                3.5s
 ✔ Container mexi-banco-v06-cuenta-2 Stopped                3.6s
 ✔ Container mexi-banco-v06-cuenta-1 Stopped                3.6s
HTTP/1.1 504 Gateway Timeout
Content-Type: application/json
Content-Length: 141

{"timestamp":"2026-10-17T07:29:19.188+00:00","path":"/api/transferencias","status":504,"error":"Gateway Timeout","requestId":"95dfe210-1295"}[1]  + 280948 done       curl -i -X POST http://localhost:8080/api/tra
nsferencias -H  -d
→  mexi-banco-s08 git:(03-gateway-discovery-balanceo) ✗ curl -s http://localhost:8080/api/cuentas/$ORIGEN
curl -s http://localhost:8080/api/cuentas/$DESTINO
{"timestamp":"2026-10-17T07:29:32.651+00:00","path":"/api/cuentas/S08O1792217021","status":503,"error":"Service Unavailable","requestId":"b23be375-1301"}{"timestamp":"2026-10-17T07:29:32.710+00:00","path":"/api/
cuentas/S08D1792217021","status":503,"error":"Service Unavailable","requestId":"0478e17b-1302"}%
→  mexi-banco-s08 git:(03-gateway-discovery-balanceo) ✗ 
```

Teniendo aquí evidencia del funcionamiento se logra ver que maneja un estado de error.

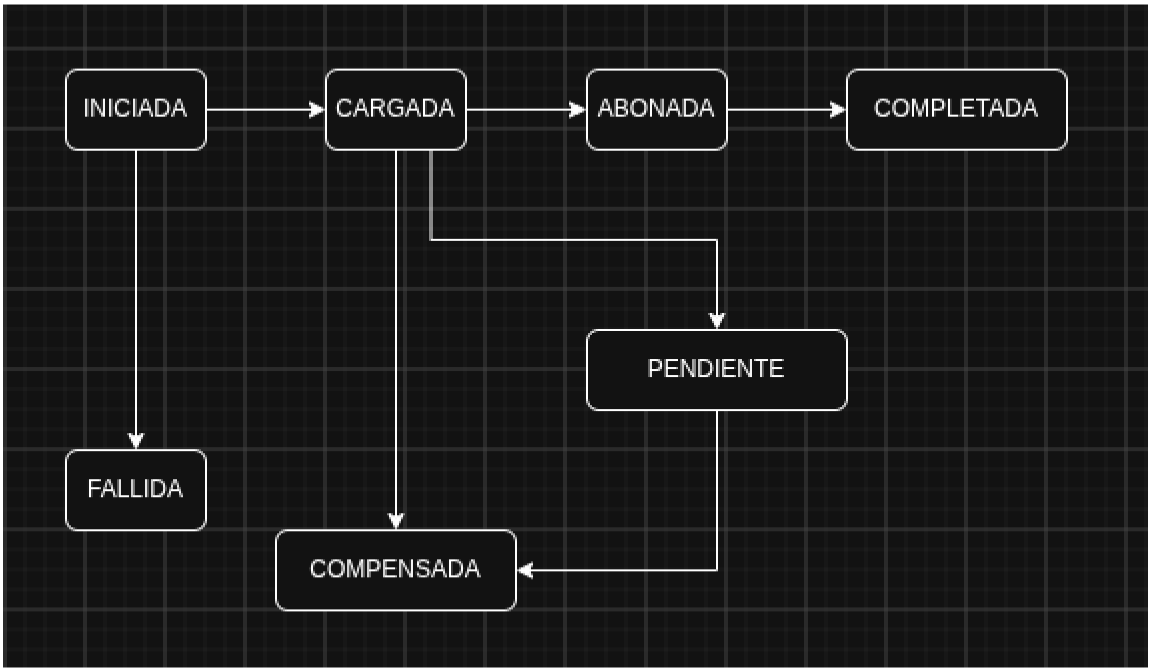
Erick Nava Santiago
Sistemas Distribuidos - Lab 08

Ejecutando prueba para los diferentes caminos:



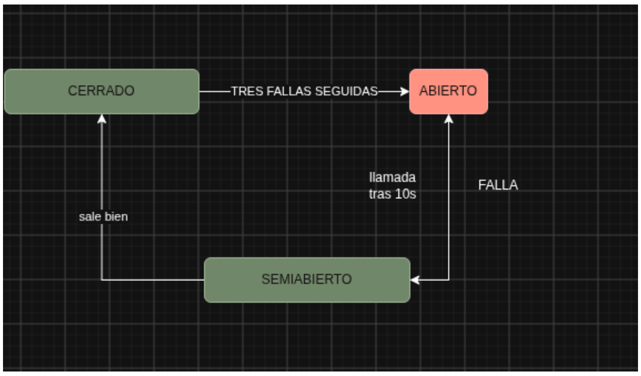
```
curl -s http://localhost:8080/api/cuentas/$ORIGEN\; echo
curl -s http://localhost:8080/api/cuentas/$DESTINO\; echo
NAME                     IMAGE                    COMMAND                SERVICE   CREATED       STATUS                   PORTS
mexi-banco-v06-cuenta-1   mexi-banco-v06-cuenta   "java -jar app.jar"   cuenta    2 hours ago   Up 4 minutes (healthy)   8080/tcp
mexi-banco-v06-cuenta-2   mexi-banco-v06-cuenta   "java -jar app.jar"   cuenta    2 hours ago   Up 4 minutes (healthy)   8080/tcp
mexi-banco-v06-cuenta-3   mexi-banco-v06-cuenta   "java -jar app.jar"   cuenta    2 hours ago   Up 4 minutes (healthy)   8080/tcp
{"id":5,"clabe":"S08O1792217021","titular":"Origen","saldo":800.00}{"id":6,"clabe":"S08D1792217021","titular":"Destino","saldo":0.00}%
→  mexi-banco-s08 git:(03-gateway-discovery-balanceo) ✗ docker compose up -d --wait transferencia

A="S08A$(date +%s)"
B="S08B$(date +%s)"
curl -s -X POST http://localhost:8080/api/cuentas -H 'Content-Type: application/json' \
  -d "{\"clabe\":\"$A\",\"titular\":\"A\",\"saldoInicial\":1000}"; echo
curl -s -X POST http://localhost:8080/api/cuentas -H 'Content-Type: application/json' \
  -d "{\"clabe\":\"$B\",\"titular\":\"B\",\"saldoInicial\":0}"; echo
[+] up 3/3
 ✓ Container mexi-banco-v06-discovery-1     Healthy                                                                                                                          5.1s
 ✓ Container mexi-banco-v06-db-1            Healthy                                                                                                                          5.1s
 ✓ Container mexi-banco-v06-transferencia-1 Healthy                                                                                                                         20.6s
{"id":7,"clabe":"S08A1792222748","titular":"A","saldo":1000}
{"id":8,"clabe":"S08B1792222748","titular":"B","saldo":0}
→  mexi-banco-s08 git:(03-gateway-discovery-balanceo) ✗ RESP=$(curl -s -w '\nHTTP %{http_code}\n' -X POST http://localhost:8080/api/transferencias \
  -H 'Content-Type: application/json' \
  -d "{\"claveOrigen\":\"$A\",\"claveDestino\":\"$B\",\"monto\":100}")
echo "$RESP"

ID1=$(echo "$RESP" | grep -o '"id":[0-9]*' | head -1 | cut -d: -f2)
curl -s http://localhost:8080/api/transferencias/$ID1\; echo
curl -s http://localhost:8080/api/cuentas/$A\; echo
curl -s http://localhost:8080/api/cuentas/$B\; echo
{"id":3,"claveOrigen":"S08A1792222748","claveDestino":"S08B1792222748","monto":100,"fecha":"2026-10-17T07:39:20.210369042Z","estado":"COMPLETADA","detalle":null}
HTTP 201
{"id":3,"claveOrigen":"S08A1792222748","claveDestino":"S08B1792222748","monto":100.00,"fecha":"2026-10-17T07:39:20.210369Z","estado":"COMPLETADA","detalle":null}{"id":7,"clabe":"S08A1792222748","titular":"A","sa
ldo":900.00}{"id":8,"clabe":"S08B1792222748","titular":"B","saldo":100.00}%
→  mexi-banco-s08 git:(03-gateway-discovery-balanceo) ✗ FALLA_ABONO_CLABE=$B docker compose up -d --wait transferencia

RESP=$(curl -s -w '\nHTTP %{http_code}\n' -X POST http://localhost:8080/api/transferencias \
  -H 'Content-Type: application/json' \
  -d "{\"claveOrigen\":\"$A\",\"claveDestino\":\"$B\",\"monto\":100}")
echo "$RESP"

ID2=$(echo "$RESP" | grep -o '"id":[0-9]*' | head -1 | cut -d: -f2)
curl -s http://localhost:8080/api/transferencias/$ID2\; echo
curl -s http://localhost:8080/api/cuentas/$A\; echo
curl -s http://localhost:8080/api/cuentas/$B\; echo
[+] up 3/3
 ✓ Container mexi-banco-v06-discovery-1     Healthy                                                                                                                          5.3s
 ✓ Container mexi-banco-v06-db-1            Healthy                                                                                                                          5.3s
 ✓ Container mexi-banco-v06-transferencia-1 Healthy                                                                                                                         20.3s
{"timestamp":"2026-10-17T07:39:53.851+00:00","path":"/api/transferencias","status":503,"error":"Service Unavailable","requestId":"cafba503-1441"}
HTTP 503
{"timestamp":"2026-10-17T07:39:53.884+00:00","path":"/api/transferencias/;","status":404,"error":"Not Found","requestId":"4eac3ceb-1442"}{"id":7,"clabe":"S08A1792222748","titular":"A","saldo":900.00}{"id":8,"cla
be":"S08B1792222748","titular":"B","saldo":100.00}%
```

Diagrama de estados:



Erick Nava Santiago
Sistemas Distribuidos - Lab 08

Diagrama para el circuit breaker:



Se agregó la carpeta resiliencia donde vivira el circuit breaker. Mismo el cual:

```
/**
* Circuit breaker hecho a mano, con los tres estados:
*
*   CERRADO     las llamadas pasas
*   ABIERTO      Se rechaza toda llamada al instante, sin intentarla
(no golpea al servicio caido ni
*                deja al cliente colgado).
*   SEMIABIERTO  Se deja pasar una sola llamada de prueba
*
*/
public class CircuitBreaker {

    public enum Estado { CERRADO, ABIERTO, SEMIABIERTO }

    /** Se lanza cuando el circuito esta abierto y la llamada se rechaza
sin intentarla. */
    public static class CircuitoAbiertoException extends
RuntimeException {
        public CircuitoAbiertoException(String mensaje) {
            super(mensaje);
        }
    }
```

```
/** Ejecuta la llamada protegida por el circuito. . */
    public <T> T ejecutar(Supplier<T> llamada) {
        boolean esPrueba = pedirPaso();
        try {
            T resultado = llamada.get();
            registrarExito(esPrueba);
            return resultado;
        } catch (Throwable error) {
            if (esFalla(error)) {
                registrarFalla(esPrueba);
            } else {
                registrarExito(esPrueba); // error de negocio: el
servicio si contesto
            }
            throw error;
        }
    }
```

Se agregaron estos dos métodos en el controller:

```

    @GetMapping("/circuito")
    public Map<String, Object> circuito() {
        return transferencias.estadoDelCircuito();
    }

    /** Circuito abierto: cuenta esta caida y la llamada ni se intento.
Es un 503, no un 500. */
    @ExceptionHandler(CircuitoAbiertoException.class)
    public ResponseEntity<Map<String, Object>>
circuitoAbierto(CircuitoAbiertoException excepcion) {
        return
ResponseEntity.status(HttpStatus.SERVICE_UNAVAILABLE).body(Map.of(
            "status", 503,
            "error", "Service Unavailable",
            "mensaje", excepcion.getMessage()));
    }
```

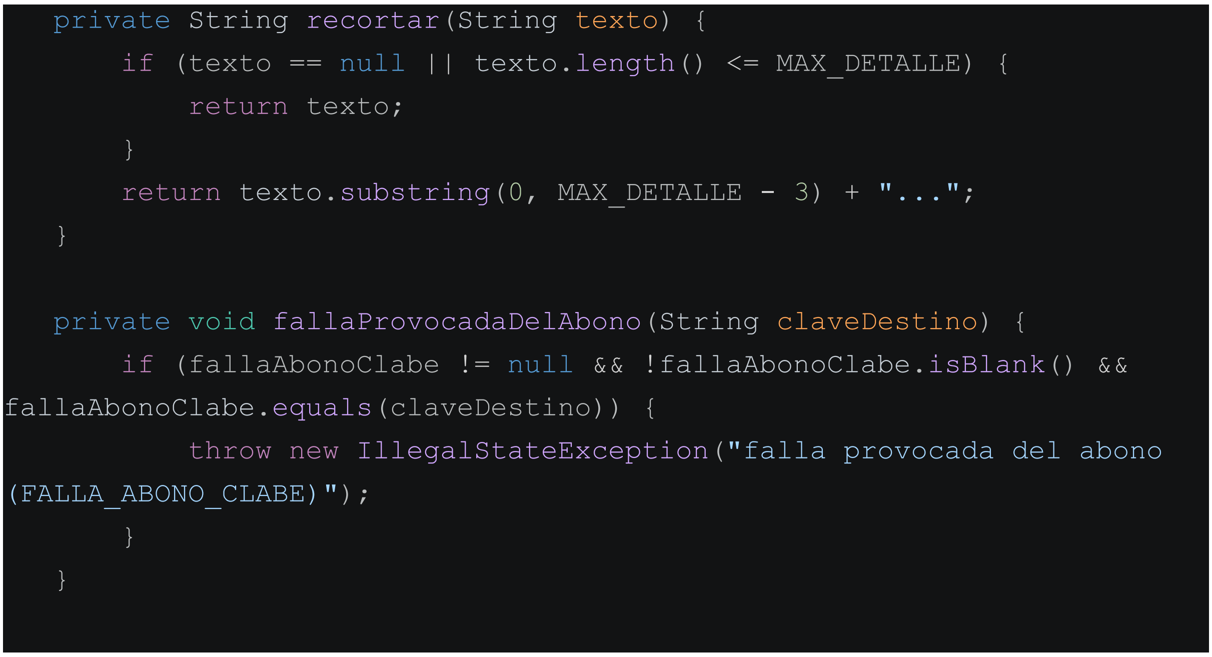
Y en Service se agregó:

```

    private final CircuitBreaker circuitoCuenta =
        new CircuitBreaker("cuenta", 3, Duration.ofSeconds(10),
TransferenciaService::esFallaDeCuenta);
```

y luego:

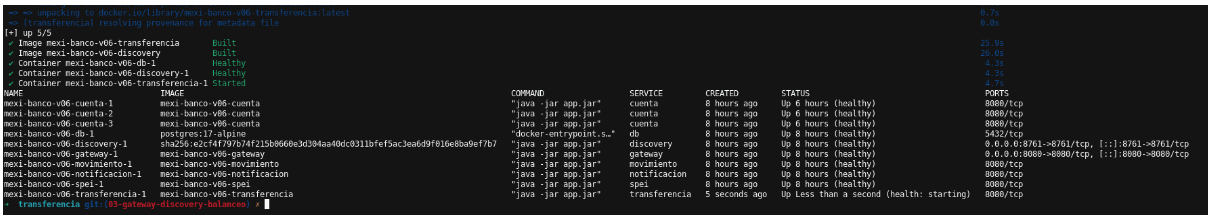
```
    /** Cambia el estado y lo confirma en la base de inmediato. */
    private void guardar(Transferencia transferencia,
EstadoTransferencia estado, String detalle) {
        transferencia.cambiarEstado(estado, recortar(detalle));
        transferencias.save(transferencia);
    }

```

Erick Nava Santiago
Sistemas Distribuidos - Lab 08



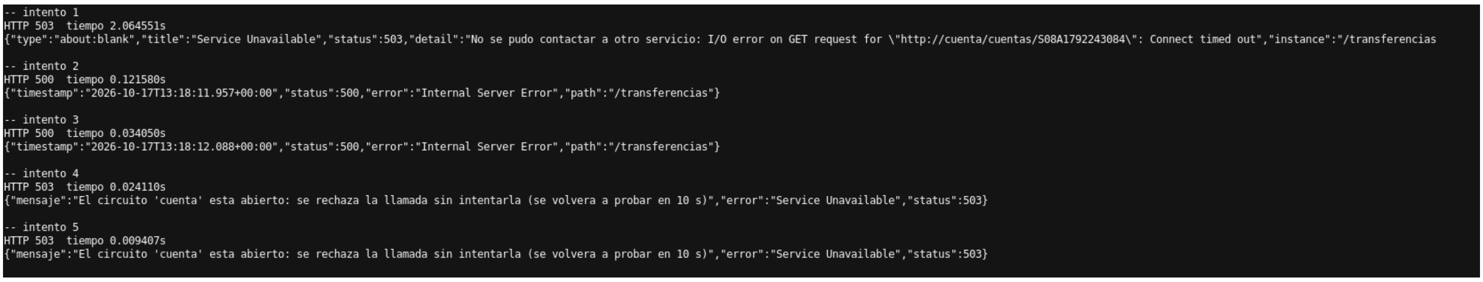
```
    private String recortar(String texto) {
        if (texto == null || texto.length() <= MAX_DETALLE) {
            return texto;
        }
        return texto.substring(0, MAX_DETALLE - 3) + "...";
    }

    private void fallaProvocadaDelAbono(String claveDestino) {
        if (fallaAbonoClabe != null && !fallaAbonoClabe.isBlank() &&
fallaAbonoClabe.equals(claveDestino)) {
            throw new IllegalStateException("falla provocada del abono
(FALLA_ABONO_CLABE)");
        }
    }
```

Se ejecutó con éxito

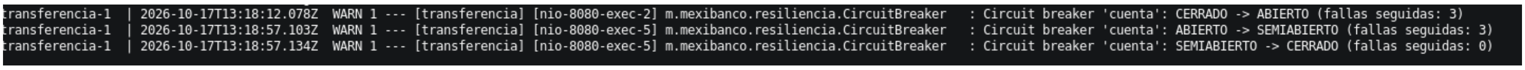


```
 => => unpacking to docker.io/library/mexi-banco-v06-transferencia:latest                                                                                              0.7s
 => [transferencia] resolving provenance for metadata file                                                                                                             0.0s
[+] up 5/5
 ✔ Image mexi-banco-v06-transferencia      Built                                                                                                                      25.9s
 ✔ Image mexi-banco-v06-discovery          Built                                                                                                                      26.0s
 ✔ Container mexi-banco-v06-db-1           Healthy                                                                                                                     4.3s
 ✔ Container mexi-banco-v06-discovery-1    Healthy                                                                                                                     4.3s
 ✔ Container mexi-banco-v06-transferencia-1 Started                                                                                                                    4.7s
NAME                             IMAGE                                                                     COMMAND                  SERVICE         CREATED         STATUS                                    PORTS
mexi-banco-v06-cuenta-1          mexi-banco-v06-cuenta                                                     "java -jar app.jar"      cuenta          8 hours ago     Up 6 hours (healthy)                      8080/tcp
mexi-banco-v06-cuenta-2          mexi-banco-v06-cuenta                                                     "java -jar app.jar"      cuenta          8 hours ago     Up 6 hours (healthy)                      8080/tcp
mexi-banco-v06-cuenta-3          mexi-banco-v06-cuenta                                                     "java -jar app.jar"      cuenta          8 hours ago     Up 6 hours (healthy)                      8080/tcp
mexi-banco-v06-db-1              postgres:17-alpine                                                        "docker-entrypoint.s…"   db              8 hours ago     Up 8 hours (healthy)                      5432/tcp
mexi-banco-v06-discovery-1       sha256:e2cf4f797b74f215b0660e3d304aa40dc0311bfef5ac3ea6d9f016e8ba9ef7b7   "java -jar app.jar"      discovery       8 hours ago     Up 8 hours (healthy)                      0.0.0.0:8761->8761/tcp, [::]:8761->8761/tcp
mexi-banco-v06-gateway-1         mexi-banco-v06-gateway                                                    "java -jar app.jar"      gateway         8 hours ago     Up 8 hours (healthy)                      0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp
mexi-banco-v06-movimiento-1      mexi-banco-v06-movimiento                                                 "java -jar app.jar"      movimiento      8 hours ago     Up 8 hours (healthy)                      8080/tcp
mexi-banco-v06-notificacion-1    mexi-banco-v06-notificacion                                               "java -jar app.jar"      notificacion    8 hours ago     Up 8 hours (healthy)                      8080/tcp
mexi-banco-v06-spei-1            mexi-banco-v06-spei                                                       "java -jar app.jar"      spei            8 hours ago     Up 8 hours (healthy)                      8080/tcp
mexi-banco-v06-transferencia-1   mexi-banco-v06-transferencia                                              "java -jar app.jar"      transferencia   5 seconds ago   Up Less than a second (health: starting)   8080/tcp
➜  transferencia git:(03-gateway-discovery-balanceo) ✗ 
```

Prueba:
Se intentaron 5 pruebas



```
-- intento 1
HTTP 503  tiempo 2.064551s
{"type":"about:blank","title":"Service Unavailable","status":503,"detail":"No se pudo contactar a otro servicio: I/O error on GET request for \"http://cuenta/cuentas/S08A1792243084\": Connect timed out","instance":"/transferencias

-- intento 2
HTTP 500  tiempo 0.121580s
{"timestamp":"2026-10-17T13:18:11.957+00:00","status":500,"error":"Internal Server Error","path":"/transferencias"}

-- intento 3
HTTP 500  tiempo 0.034050s
{"timestamp":"2026-10-17T13:18:12.088+00:00","status":500,"error":"Internal Server Error","path":"/transferencias"}

-- intento 4
HTTP 503  tiempo 0.024110s
{"mensaje":"El circuito 'cuenta' esta abierto: se rechaza la llamada sin intentarla (se volvera a probar en 10 s)","error":"Service Unavailable","status":503}

-- intento 5
HTTP 503  tiempo 0.009407s
{"mensaje":"El circuito 'cuenta' esta abierto: se rechaza la llamada sin intentarla (se volvera a probar en 10 s)","error":"Service Unavailable","status":503}
```



```
transferencia-1  | 2026-10-17T13:18:12.078Z  WARN 1 --- [transferencia] [nio-8080-exec-2] m.mexibanco.resiliencia.CircuitBreaker   : Circuit breaker 'cuenta': CERRADO -> ABIERTO (fallas seguidas: 3)
transferencia-1  | 2026-10-17T13:18:57.103Z  WARN 1 --- [transferencia] [nio-8080-exec-5] m.mexibanco.resiliencia.CircuitBreaker   : Circuit breaker 'cuenta': ABIERTO -> SEMIABIERTO (fallas seguidas: 3)
transferencia-1  | 2026-10-17T13:18:57.134Z  WARN 1 --- [transferencia] [nio-8080-exec-5] m.mexibanco.resiliencia.CircuitBreaker   : Circuit breaker 'cuenta': SEMIABIERTO -> CERRADO (fallas seguidas: 0)
```

La saga garantiza que la transferencia termine completa o compensada pero no da aislamiento, lo que significa que son visibles para cualquiera. En las pruebas se lograba ver en origen 900 y en el destino 0 pese a que el dinero no estaba en ninguna de las dos. siendo esto que quien la consulte tendría una consulta fantasma, violando la continuidad de los datos, cosa que no pasaba en el monolito gracias a el aislamiento que se puede manejar en una sola pieza.